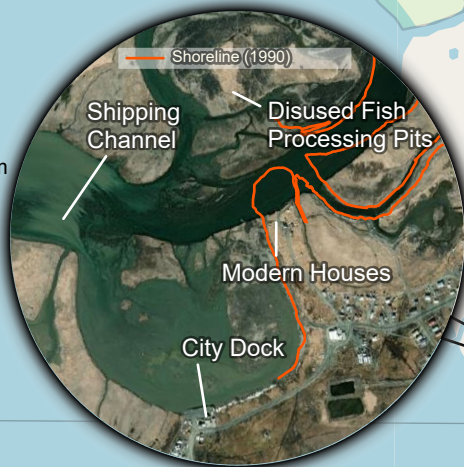




Map Projection: NAD 1983 2011 Alaska Albers
Ellipsoid: GRS80

At Focus Area 1, a peninsula with modern houses has experienced over 100m of loss since 1990. Archaeological remains in the form of historic fish processing pits are also at risk.



The shipping channel has also taken to moving erratically in recent years, jeopardizing the all-important biannual delivery of fuel and supplies that Quinhagak needs to survive.

Map of Subsistence Activity and Erosion-Based Threats Near Quinhagak, Alaska, June 2022

By Jonathan Lim (University of Oxford), Dr. Sean Gleason, and Daniel Marsden (Hampden Sydney College).

Descriptive Notes and Data Sources

This is a map of subsistence activity in the immediate vicinity of Quinhagak, an Alaska Native Yupik community of around 700 people on the coast of the Bering Sea. Climate change-driven erosion poses a threat to centuries-old traditional subsistence areas and modern infrastructure alike. This map draws data from the 1999 survey work of Joseph Pleasant (Former land manager, Qarnituuq Corp.), a 2005 map entitled "Area Use Map Quinhagak" by the Coastal Villages Region Fund, a 1990 aerial image of uncertain provenance, and the ethnographic interviews of the authors from 2019-2021. The inset satellite imagery, hosted on ESRI Living Atlas, dates from 2019.

Map data © OpenStreetMap contributors, Microsoft, Facebook, Inc. and its affiliates, Esri Community Maps contributors, Map layer by Esri, Airbus, USGS, NGA, NASA, CGIAR, NCEAS, NLS, OS, NMA, Geodastystyrelsen, GSA, GSI and the GIS User Community, Esri, USGS, Esri, Garmin, FAO, NOAA, USGS, Maxar

The Qanirtuuq is the lifeblood of Quinhagak. It brings salmon to the village every summer and is an important waterway for accessing vital subsistence resources further inland.

Qanirtuuq (Kanektok River)

Gravel Pits

Quinhagak Airport

QUINHAGAK

Focus Area 2: Meqsarturyaraq

Focus Area 1: Kuiggacuk

Old Airfield

Water intake valve

Old Airfield Road

At Focus Area 2, rapid erosion has led to the abandonment of Quinhagak's old airfield in 2004. More than 40m of land has been eroded since 2005. The erosion here is devastating, as it threatens residential areas and cuts off access to important fishing areas and gravel harvesting pits to the east.

The area around a water intake valve is also at risk and may need to be dismantled soon, endangering the community's fresh water supply and sewage system.

Legend

- Areas of Concern
- Native Allotments
- Bird Hunting
- Fish Camps and Fishing Activity
- Small Mammal Hunting
- Vegetation Harvesting
- Fish Racks (2005)
- Fish Processing Area (2005)
- Berry Harvesting Areas (2005)

0 0.5 1 2 km



-161°55'

-161°50'